

BONILLA ISLAND, BC, Canada

WMO: 714840

Lat: 53.493N

Lon: 130.637W

Elev: 13

StdP: 101.17

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-4.2	-2.0	-15.7	1.0	-2.0	-12.6	1.3	0.0	23.8	5.6	21.1	6.3	4.0	80	0.839

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.4	16.4	13.8	15.5	13.6	14.8	13.3	14.6	15.5	14.1	15.0	13.7	14.6	4.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.1	10.1	15.2	13.6	9.7	14.7	13.2	9.5	14.3	41.0	15.7	39.5	15.1	38.5	14.7	18.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.6	17.8	15.5	DB	-6.4	19.3	3.4	2.5	-8.9	21.1	-10.9	22.6	-12.9	24.0	-15.4	25.8	(4)
(5)			WB	-9.9	16.3	4.0	0.9	-12.8	17.0	-15.2	17.5	-17.5	18.1	-20.4	18.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	5.2	4.9	5.3	7.1	9.1	11.2	12.8	13.5	12.4	9.8	7.0	5.2		
	DBStd	3.74	3.14	2.45	2.44	1.72	1.51	1.32	1.22	1.11	1.23	1.62	2.64	2.84		
	HDD10.0	833	150	143	146	90	36	3	0	0	1	22	92	150		
	HDD18.3	3539	408	376	404	338	287	215	170	151	178	264	339	408		
	CDD10.0	334	1	0	1	2	7	38	88	107	72	16	3	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	7.0	7.8	7.4	7.5	7.3	6.8	6.3	5.9	5.8	6.3	7.6	7.8	7.9		
(15) Precipitation	PrecAvg	2169	216	193	183	166	123	102	88	111	175	288	263	246		
	PrecMax	2620	318	385	339	329	230	179	201	267	330	472	457	359		
	PrecMin	1724	63	21	42	49	31	16	26	14	53	130	110	104		
	PrecStd	255	66	88	73	74	57	44	54	59	66	77	89	65		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	11.2	12.3	14.4	15.2	16.8	17.0	17.4	17.3	14.2	14.9	11.3		
		MCWB	7.1	6.7	7.0	8.9	9.6	13.5	14.5	15.1	12.6	10.5	12.5	8.0		
	2%	DB	10.0	9.5	9.7	11.1	12.4	14.4	15.7	16.3	15.3	13.0	11.6	9.9		
		MCWB	7.6	7.1	7.5	8.6	10.1	12.4	14.0	14.5	13.1	11.4	9.3	8.1		
	5%	DB	9.0	8.6	8.7	9.7	11.5	13.4	15.0	15.6	14.5	12.3	10.7	9.1		
		MCWB	7.8	6.9	7.2	8.1	9.7	11.8	13.5	14.0	12.9	10.8	9.1	7.7		
	10%	DB	8.4	7.8	8.1	9.0	10.8	12.7	14.4	15.0	14.0	11.8	10.1	8.5		
		MCWB	7.2	6.4	6.7	7.5	9.3	11.3	13.1	13.6	12.7	10.4	8.7	7.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	8.9	9.0	9.9	11.7	13.7	15.1	15.6	14.5	12.9	14.8	9.5		
		MCDB	9.7	9.6	10.2	11.8	13.1	15.1	16.2	16.6	15.1	13.3	15.8	10.0		
	2%	WB	8.5	8.2	8.3	9.2	10.6	12.7	14.3	14.8	13.9	12.1	10.5	8.7		
		MCDB	9.1	9.0	9.5	10.5	11.8	13.8	15.3	15.7	14.5	12.7	11.1	9.4		
	5%	WB	8.0	7.6	7.8	8.6	10.1	12.1	13.9	14.3	13.5	11.4	9.9	8.0		
		MCDB	8.7	8.4	8.9	9.7	11.0	13.1	14.8	15.2	14.2	12.2	10.6	8.8		
	10%	WB	7.6	7.0	7.2	8.0	9.6	11.6	13.4	13.9	13.1	10.8	9.1	7.4		
		MCDB	8.4	7.9	8.3	9.1	10.6	12.6	14.4	14.8	13.9	11.7	10.1	8.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	2.9	3.1	3.1	2.8	2.4	2.3	2.2	2.4	2.3	2.5	3.0	3.0		
		MCDBR	3.1	3.7	3.5	3.9	3.4	3.4	3.0	3.3	3.2	2.8	3.2	3.1		
	5% WB	MCWBR	2.5	2.9	2.4	2.5	2.4	2.2	2.1	2.4	2.2	2.4	3.0	2.8		
		MCWBR	2.6	3.2	3.1	3.5	3.0	3.0	2.9	3.0	2.6	2.4	3.1	3.0		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.319	0.344	0.380	0.376	0.375	0.369	0.370	0.363	0.345	0.325	0.304		
	taud		2.411	2.499	2.439	2.324	2.402	2.440	2.499	2.511	2.496	2.451	2.336	2.339		
	Ebn at Noon		668	796	842	849	873	876	875	855	813	743	627	589		
	Edh at Noon		55	69	89	113	110	107	99	93	84	71	58	49		
(44) All-Sky Solar Radiation	RadAvg		0.56	1.40	2.53	3.97	5.00	5.22	4.92	4.19	2.83	1.59	0.75	0.46		
	RadStd		0.07	0.22	0.28	0.21	0.42	0.37	0.51	0.32	0.29	0.20	0.08	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	+0.38	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page